

Database Design & Development

Database System for FoodHub
Company

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Unit 04 – Database Design &
Development
03/17/2026



Higher Nationals - Summative Assignment Feedback Form

Student Name/ID			
Unit Title	Unit 04 – Database Design & Development		
Assignment Number	1 of 2	Assessor	
Submission Date		Date Received 1st submission	
Re-submission Date		Date Received 2nd submission	

Assessor Feedback:

LO1 Use an appropriate design tool to design a relational database system for a substantial problem ☐

Pass, Merit & Distinction ☐ P1 M1 D1 ☐ ☐

Descripts

LO2 Create a Design document for the total system of the given scenario.

Pass, Merit & Distinction ☐ P2 M2 D1 ☐ ☐

Descripts

* Please note that grade decisions are provisional. They are only confirmed once internal and external moderation has taken place and grades decisions have been agreed at the assessment board.

Assessor Feedback:

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Grade:	Assessor Signature:	Date:
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Programme:			
Unit: Unit 04			
Assignment number and title: Database System for FoodHub Company			

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Student Declaration

Student declaration	
I certify that the assignment submission is entirely my own work and I fully understand the consequences of plagiarism. I understand that making a false declaration is a form of malpractice.	
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Unit 04: Database Design & Development

Assignment Brief

Student Name/ID Number	
Unit Number and Title	Unit 04 – Database Design & Development
Academic Year	
Unit Tutor	
Assignment Title	Database System for FoodHub Company
Issue Date	
Submission Date	
Submission Format	
The assignment submission is in the form of the following. • The submission should be in the form of an individual written report written in a concise, formal business style using single spacing and font size 12. • This document should include all the tasks given in the brief with the relevant design diagrams. • A conceptual relational database. This should contain the all types of required designs of the total system according to the given scenario. • The recommended word limit is 3,000 – 3,500 words, although you will not be penalised for exceeding the total word limit. All work must be supported with research and referenced using the Harvard referencing system	
Unit Learning Outcomes	
LO1	Use an appropriate design tool to design a relational database system for a substantial problem
LO2	Create a Design document for the total system of the given scenario.

Transferable skills and competencies developed

Computing-related cognitive skills :

- Demonstrate knowledge and understanding of essential facts, concepts, principles and theories relating to computing and computer applications
- Use such knowledge and understanding in the modelling and design of computer-based systems for the purposes of comprehension, communication, prediction and the understanding of trade-offs
- Recognise and analyse criteria and specifications appropriate to specific problems, and plan strategies for their solutions

Computing-related practical skills :

- The ability to specify, design and construct reliable, secure and usable computer-based systems
- The ability to evaluate systems in terms of quality attributes and possible trade-offs presented within the given problem
- The ability to critically evaluate and analyse complex problems, including those with incomplete information, and devise appropriate solutions, within the constraints of a budget.

Generic skills for employability:

- Intellectual skills: critical thinking; making a case; numeracy and literacy
- Self-management: self-awareness and reflection; goal setting and action planning
- Independence and adaptability; acting on initiative; innovation and creativity
- Contextual awareness, e.g. the ability to understand and meet the needs of individuals, business and the community, and to understand how workplaces and organisations are governed.

Vocational scenario

FoodHub Company is a well-known fast food company located in the United States of America. It was started 14 years ago & the company management decided to open branches worldwide. So they are going to open a new branch in Sri Lanka as well. The company decided to hire a local database expert to develop a Database Management system to assist in this project for decision making. Therefore, the data should maintain accuracy & the consistency throughout the data life cycle.

FoodHub Company only delivering foods for the customers & they are not offer or selling food in the outlets. For delivery they have separate staff, who are known as **Riders**. There are other type of employees as well, but their details are not recorded in the database in the initial level. In the initial level the company consider to keep records of food deliveries & related information only. If the system success, they will enhanced for Human Resource Management & Finance Management integrate with initial system of Delivery Management system.

In this system the Customer will identify with unique Customer ID, Customer Name, NIC, Date of Birth, Contact Number & Address with Location No., Lane, Street & city. The customers' address will consider as the order delivery address. One customer can place many orders. But one order will have only one customer. The order can be place over the phone & it will enter by a staff member of the company. Orders will identify by unique Order No, Order Date, Order Time, Order Status, Payment method that customer going to pay & order amount according to the order items. One order can have many items & one food item can be available in many orders. Items can be identify by Item Number, item name, Item Category & Price. Company facilitate the customers to know what the ingredients of the particular food item are. For that food items will contain ingredients as well, every food of the company will make with minimally with three ingredients. If the customer needs to know the ingredients of a particular food item, they can request while placing the order. The staff member who takes the order will able to provide the ingredients when it available in the system. When the order is dispatch from the outlet, the dispatched time should record in the database.

In FoodHub company all food orders will be deliver by **Riders** who are identify by Employee number, Employee name with First Name, Middle Name & Last Name, NIC, Date of Birth, Contact Number, License number, Address & Age in the system. The company keeps records about the dependents of the employees as well. But in the initial level of the project they are willing to keep only the dependents records of the riders. Dependents will be identify by the dependent name, the relationship to the rider & Date of Birth. But they don't have their own unique field, so the company expect a solution for this problem as well. So as the database administrator you have to provide the most appropriate solution according to the theories & techniques of the relational database.

Each rider will deliver the food items for the customers by using the company motor bikes & they are not assign to a specific rider. When they attend to work according to their shift

the available motor bike will be assign to the rider for that shift. The motor bikes will be identify by unique vehicle registration number, brand, model, engine number, vehicle registered date & the colours of the particular motorbike. None of the motor bike is coloured with a single colour, the company use different themes to colour the bikes. According to the database theories you have to provide the facility to identify the records of that which bike is used by which rider on a specific date. Also need to keep records of the motor bike meter readings when it assigned to the rider & when it return by the rider. In the System development level the company doesn't need the report generated part initially, it will include in the next level of the development. But the Database designs & system designs should include the reporting processes.

Assignment activity and guidance

Based on FoodHub Company scenario you will need to:

1. Use an appropriate design tool to design a relational database system for a substantial problem
 - a. Identify the user and system requirements of the FoodHub Company System.
 - b. According to the above requirements that identified, design a database and design a relational database system using conceptual design (ER Model) with including identifiers (primary Key) of entities and attributes, cardinalities, participations of relationships.
 - c. Convert the ER Model into logical database design using relational database model including primary keys foreign keys and referential Integrities. It should contain at least six interrelated tables.
 - d. Check whether the provided logical design is normalised. If not, normalize the database by removing the anomalies.

Note: It is allowed to have your own appropriate assumptions and related attributes within the scope of the case study given

2. Prepare a Design Document with the following designs
 - a. Design set of simple interfaces to input and output for the above scenario using Wireframe or any interface-designing tool. *(Those designs cannot be the interface designs of visual studio.)*

- b. Design Flow charts for 04 major processes of the given scenario. Those 04 major processes can be select from the same module of the system **or** different modules of the system according to the student's preference.
- c. Design a Use Case Diagram for the given scenario with including all available processes of the system.
- d. According to the scenario design a class diagram with considering the basic requirements of the system. (No need to mention the access levels of the class attributes)
- e. Design a Data Flow Diagrams with Context Level & Level 01, to demonstrate the flow of data within the system. Use standard notations of Gane & Sarson model & each level should contain the correct components as well.
- f. Evaluate the effectiveness of the given designs (Each & every design should be evaluated) in terms of the identified user and system requirements.

(** Create a design document inside of the same submission document for the above a,b,c,d & e activities.)

Recommended Resources

Please note that the resources listed are examples for you to use as a starting point in your research – the list is not definitive.

Weblinks:

<https://support.microsoft.com/en-GB> (2022) Database design basics [online] Available at: <https://support.microsoft.com/en-us/office/database-design-basics-eb2159cf-1e30-401a-8084-bd4f9c9ca1f5> [Accessed 1 August 2022]

<https://www.guru99.com/> (2022) Database Design in DBMS Tutorial: Learn Data Modeling [online] Available at: <https://www.guru99.com/database-design.html> [Accessed 1 August 2022]

<https://www.integrate.io/> (2021) Complete Guide to Database Schema Design [online] Available at: <https://www.integrate.io/blog/complete-guide-to-database-schema-designguide/> [Accessed 1 August 2022]

<https://www.lucidchart.com/pages/> (2022) Database Structure and Design Tutorial [online] Available at: <https://www.lucidchart.com/pages/database-diagram/database-design> [Accessed 1 August 2022]

Journal articles:

Batra, D. & Davis, J. (1992). Conceptual data modelling in database design: similarities and differences between expert and novice designers. International Journal of Man-Machine Studies, Volume 37, Issue 1, 1992, pp. 83-101. [https://doi.org/10.1016/00207373\(92\)90092-Y](https://doi.org/10.1016/00207373(92)90092-Y).

Gunjal, B. (2003). Database System: Concepts and Design. Proceedings of 24th IASLIC–SIG-2003.

Kaur, T. & Singh B. (2003). Testing of Databases. IJSET - International Journal of Innovative Science, Engineering & Technology, Vol. 1 Issue 6. ISSN 2348 – 7968

Kaur, Taranpreet & Sehra, Sumeet Kaur. (2015). Designing and Development of Database Testing Tool. International Journal of Computer Applications (0975 – 8887) Volume 120 – No.19. 14. doi:10.5120/21334-4330.

Kraleva, Radoslava & Kralev, Velin & Sinyagina, Nina & Koprinkova-Hristova, Petia & Bocheva, Nadejda. (2018). Design and Analysis of a Relational Database for Behavioral Experiments Data Processing. International Journal of Online Engineering (iJOE). 14. 117. doi:10.3991/ijoe.v14i02.7988.

Letkowski, J. (2015). Doing database design with MySQL. Journal of Technology Research. Volume 6.

Reading:

Captain, F. (2013) Six-Step Relational Database Design™: A step by step approach to relational database design and development, 2nd edn, CreateSpace Independent Publishing Platform

Hernandez, M. (2003) Database Design for Mere Mortals: A Hands-On Guide to Relational Database Design, 2nd edn, Addison Wesley

Stephens, R. (2008) Begin Database Design W / WS (Wrox Programmer to Programmer), 1st edn, Jossey-Bass

HN Global:

HN Global HN Global (2021) Reading Lists. Available at:

<https://hnglobal.highernationals.com/learning-zone/reading-lists>

HN Global (2021) Student Resource Library. Available at:

<https://hnglobal.highernationals.com/subjects/resource-libraries>

HN Global (2021) Textbooks. Available at:

<https://hnglobal.highernationals.com/textbooks>

Learning Outcomes and Assessment Criteria

Pass	Merit	Distinction
LO1 Use an appropriate design tool to design a relational database system for a substantial problem		D1 Evaluate the effectiveness of the design in relation to user and system requirements. Provide Database Schema in Textual format & Graphical format.
P1 Design a relational database system using appropriate design tools & techniques, containing at least six interrelated tables, with clear statements of user and system requirements	M1 Produce a comprehensive design for a fully-functional system, which includes interface and output designs & data normalisation.	
LO2 Create a Design document for the total system of the given scenario.		
P2 Create Design document for the given scenario with all required design techniques.	M2 Each & every design with appropriate notations & Before the Design document, explain the design concepts.	

INTRODUCTION

This report presents the design and development of a relational Database Management System for FoodHub Company, a food delivery organisation expanding into Sri Lanka. The company currently lacks a structured database system, making it difficult to efficiently manage customer records, food orders, rider details, and delivery assignments. The proposed system aims to address these challenges by providing an accurate, consistent, and well-organised database solution to support the company's day-to-day operations and decision-making processes.

The primary objective of this system is to design a normalised relational database containing a minimum of six interrelated tables, covering customers, orders, food items, ingredients, riders, dependents, motorbikes, and bike assignments. The design applies Entity-Relationship modelling, relational schema conversion, and normalisation up to Third Normal Form (3NF) to eliminate data redundancy and ensure referential integrity throughout the system.

This report covers the complete design process, including user and system requirements, conceptual and logical database design, normalisation, interface wireframes, flowcharts, Use Case Diagram, Class Diagram, and Data Flow Diagrams using Gane and Sarson notation. Each design artefact is evaluated against the identified requirements to demonstrate its effectiveness in meeting the operational needs of FoodHub Company.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to several individuals and organisations who have supported me throughout the preparation of this report on the Design and Development of a Relational Database Management System for FoodHub Company.

First and foremost, I would like to thank my Unit Tutor and academic staff at my institution for their invaluable guidance, encouragement, and expertise throughout this unit. Their insights and constructive feedback have been instrumental in shaping my understanding of database design principles, relational modelling, and system design methodologies.

I would also like to extend my appreciation to the Pearson BTEC Higher National programme for providing a well-structured curriculum and access to a wealth of learning resources through HN Global. The recommended reading materials, weblinks, and journal articles provided as part of Unit 04 have been particularly valuable in supporting the research and design work presented in this report.

Additionally, I am grateful to my peers and colleagues for their support and collaboration throughout this project. Their ideas, discussions, and constructive input have significantly contributed to the development of the designs and analyses presented in this report.

Lastly, I would like to express my heartfelt thanks to my friends and family for their constant encouragement and unwavering support throughout my academic journey. Their trust in my abilities has inspired me to strive for excellence in all aspects of my studies. I deeply value the advice and assistance of everyone who has contributed to the successful completion of this report.

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User and System Requirements

User Requirements

User requirements describe what the end users expect the system to accomplish from a functional perspective. Based on the FoodHub Company scenario, the following user requirements have been identified:

- The system shall allow staff members to register new customers with their personal details including Customer ID, Name, NIC, Date of Birth, Contact Number, and Address.
- The system shall enable staff members to record food orders placed by customers over the phone, capturing Order Number, Date, Time, Status, Payment Method, and Amount.
- The system shall allow staff members to view and retrieve the ingredients of any food item upon customer request during order placement.
- The system shall enable the dispatch staff to record the dispatch time when an order leaves the outlet.
- The system shall allow the system to assign available motor bikes to riders at the beginning of each shift and record the meter reading at assignment and return.
- The system shall maintain records of rider (employee) details including Employee Number, Name, NIC, Date of Birth, Contact Number, License Number, Address, and Age.
- The system shall keep records of the dependents of riders, identified by name, relationship, and date of birth.
- The system shall provide future scalability to integrate HR Management and Finance Management modules.
- The system shall support future reporting features (report generation not required at this development stage).

System Requirements

System requirements define the technical specifications that the database management system must satisfy:

- The system shall implement a relational database with a minimum of six interrelated tables.
- All primary keys shall uniquely identify records within their respective tables.
- Referential integrity shall be enforced through foreign key constraints across all related tables.
- The Customer entity shall be uniquely identified by Customer ID.
- The Order entity shall be uniquely identified by Order Number and shall link to exactly one customer, while one customer may have many orders (one-to-many relationship).

- The Order-Item relationship shall be implemented as a many-to-many associative entity (Order_Item table) since one order can contain many items and one item can appear in many orders.
- The Food Item entity shall contain ingredients; since every item has a minimum of three ingredients and one ingredient may be shared by many items, a many-to-many relationship must be modelled between Food_Item and Ingredient.
- The Rider (Employee) entity shall be uniquely identified by Employee Number.
- Rider Dependents shall be represented in a weak entity since they have no unique identifier of their own; a composite primary key of (Employee_Number, Dependent_Name, Date_of_Birth) shall be used.
- Motor bikes shall be uniquely identified by Vehicle Registration Number. Multiple colours per bike shall be modelled as a multi-valued attribute (separate table).
- The bike assignment history shall be recorded in an Assignment table capturing Rider, Bike, Assignment Date, Start Meter Reading, and End Meter Reading.
- The database shall be normalised to at least Third Normal Form (3NF) to eliminate data anomalies.
- The system shall use cascading referential integrity rules to maintain data consistency.

Conceptual Design – Entity-Relationship (ER) Model

2.1 Entities and Attributes

The following entities have been identified from the FoodHub scenario. Primary keys are underlined in the ER diagram; they are indicated with (PK) here.

Table 1:entity and attributes

Entity	Attributes	Notes
CUSTOMER	<u>Customer_ID</u> (PK), Customer_Name, NIC, Date_of_Birth, Contact_Number, Location_No, Lane, Street, City	Strong entity
ORDER	<u>Order_No</u> (PK), <u>Order_Date</u> , <u>Order_Time</u> , <u>Order_Status</u> , Payment_Method, Order_Amount,Dispatch_Time, Customer_ID (FK)	Strong entity; Dispatch_Time recorded when dispatched

FOOD_ITEM	Item_Number (PK), Item_Name, Item_Category, Price	Strong entity
ORDER_ITEM	Order_No (FK, PK), Item_Number (FK, PK), Quantity	Associative/junction table for M:N between Order and Food_Item
INGREDIENT	Ingredient_ID (PK), Ingredient_Name	Strong entity
ITEM_INGREDIENT	Item_Number (FK, PK), Ingredient_ID (FK, PK)	Associative table for M:N between Food_Item and Ingredient
RIDER (EMPLOYEE)	Employee_No (PK), First_Name, Middle_Name, Last_Name, NIC, Date_of_Birth, Contact_Number, License_Number, Address, Age	Strong entity
DEPENDENT	Employee_No (FK, PK), Dependent_Name (PK), Date_of_Birth (PK), Relationship	Weak entity – composite PK solves missing unique identifier
MOTORBIKE	Registration_No (PK), Brand, Model, Engine_Number, Registered_Date	Strong entity; colours stored separately
BIKE_COLOUR	Registration_No (FK, PK), Colour (PK)	Resolves multi-valued attribute of colours
ASSIGNMENT	Assignment_ID (PK), Employee_No (FK), Registration_No (FK), Assignment_Date, Start_Meter_Reading, End_Meter_Reading	Tracks which rider used which bike on a specific date

2.2 Relationships, Cardinalities, and Participations

Table 2: Relationship, cardinalities and participations

Relationship	Entities	Cardinality	Participation
PLACES	CUSTOMER – ORDER	One-to-Many (1: N)	Total on ORDER (every order must have a customer); Partial on CUSTOMER (a customer may not

			have placed any order yet)
CONTAINS	ORDER – FOOD_ITEM (via ORDER_ITEM)	Many-to-Many (M:N)	Total on ORDER_ITEM; Partial on both ORDER and FOOD_ITEM
MADE_OF	FOOD_ITEM – INGREDIENT (via ITEM_INGREDIENT)	Many-to-Many (M:N)	Total on FOOD_ITEM (min 3 ingredients); Partial on INGREDIENT
DELIVERS	RIDER – ORDER	One-to-Many (1:N)	Partial on both sides
HAS_DEPENDENT	RIDER – DEPENDENT	One-to-Many (1:N)	Total on DEPENDENT (weak entity); Partial on RIDER
ASSIGNED_TO	RIDER – MOTORBIKE (via ASSIGNMENT)	Many-to-Many (M:N)	Partial on both; tracks historical assignments
HAS_COLOUR	MOTORBIKE – BIKE_COLOUR	One-to-Many (1:N)	Total on BIKE_COLOUR (every bike has at least one colour)

2.3 ER Diagram (Textual Representation)

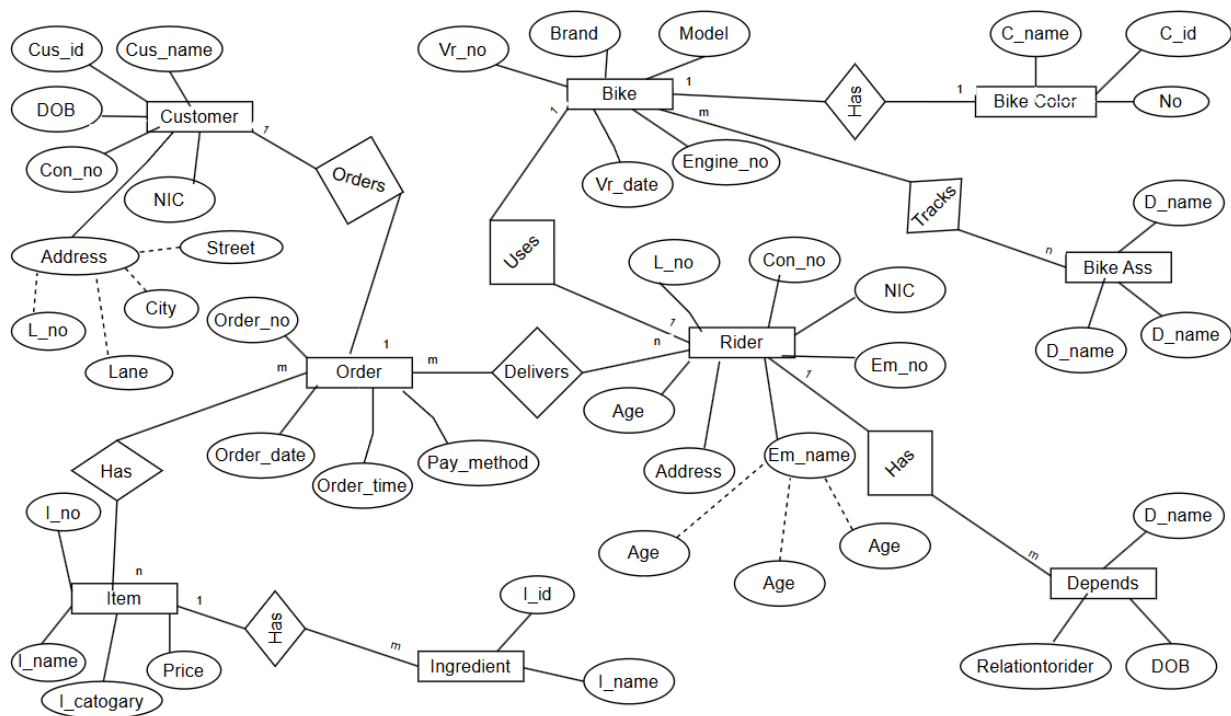


Figure 1:ER Diagram

Below is a textual description of the ER Model.

Diagram Description:

- CUSTOMER [rectangle] — < PLACES > — ORDER [rectangle] (1:N, Customer_ID is FK in ORDER)
- ORDER [rectangle] — < CONTAINS > — FOOD_ITEM [rectangle] resolved via ORDER_ITEM junction [rectangle]
- FOOD_ITEM [rectangle] — < MADE_OF > — INGREDIENT [rectangle] resolved via ITEM_INGREDIENT junction [rectangle]
- RIDER [rectangle] — < HAS_DEPENDENT > — DEPENDENT [double rectangle, weak entity] (1:N, composite PK)
- RIDER [rectangle] — < ASSIGNED_TO > — MOTORBIKE [rectangle] resolved via ASSIGNMENT junction [rectangle] with Assignment_Date, Start_Meter, End_Meter
- MOTORBIKE [rectangle] — < HAS_COLOUR > — BIKE_COLOUR [rectangle] (1:N, multi-valued attribute resolved)
- RIDER [rectangle] — < DELIVERS > — ORDER [rectangle] (1:N, Employee_No is FK in ORDER)

3. Logical Database Design – Relational Schema

The ER model is converted to a relational schema below. Primary Keys (PK) are shown in bold; Foreign Keys (FK) are indicated. All referential integrity constraints are specified.

3.1 CUSTOMER Table

Table 3:customer table

Column	Data Type	Constraint	Description
Customer_ID	VARCHAR(10)	PRIMARY KEY	Unique identifier for each customer
Customer_Name	VARCHAR(100)	NOT NULL	Full name of customer
NIC	VARCHAR(12)	UNIQUE, NOT NULL	National Identity Card number
Date_of_Birth	DATE	NOT NULL	Customer date of birth
Contact_Number	VARCHAR(15)	NOT NULL	Phone contact number
Location_No	VARCHAR(10)	NOT NULL	Location number of address
Lane	VARCHAR(50)	NULL	Lane name
Street	VARCHAR(100)	NOT NULL	Street name
City	VARCHAR(50)	NOT NULL	City of customer address

3.2 RIDER Table

Table 4:Rider table

Column	Data Type	Constraints	Description
Employee_No	VARCHAR(10)	PRIMARY KEY	Unique employee identifier
First_Name	VARCHAR(50)	NOT NULL	First name
Middle_Name	VARCHAR(50)	NULL	Middle name (optional)
Last_Name	VARCHAR(50)	NOT NULL	Last name
NIC	VARCHAR(12)	UNIQUE, NOT NULL	National Identity Card
Date_of_Birth	DATE	NOT NULL	Date of birth
Contact_Number	VARCHAR(15)	NOT NULL	Contact phone number
License_Number	VARCHAR(20)	UNIQUE, NOT NULL	Driving licence number
Address	VARCHAR(200)	NOT NULL	Full address
Age	INT	NOT NULL	Age (can be computed from DOB)

3.3 DEPENDENT Table

Table 5:dependent table

Column	Data Type	Constraint	Description
Employee_No	VARCHAR(10)	PRIMARY KEY, FK → RIDER(Employee_No)	Links to rider
Dependent_Name	VARCHAR(100)	PRIMARY KEY	Name of dependent
Date_of_Birth	DATE	PRIMARY KEY	DOB (part of composite PK)
Relationship	VARCHAR(50)	NOT NULL	Relationship to rider (e.g. Child, Spouse)

A composite primary key of (Employee_No, Dependent_Name, Date_of_Birth) uniquely identifies each dependent. This satisfies the requirement that dependents have no single unique field of their own.

3.4 MOTORBIKE Table

Table 6:motorbike table

Column	Data Type	Constraint	Description
Registration_No	VARCHAR(15)	PRIMARY KEY	Unique vehicle registration
Brand	VARCHAR(50)	NOT NULL	Motorbike brand (e.g. Honda)
Model	VARCHAR(50)	NOT NULL	Motorbike model
Engine_Number	VARCHAR(20)	UNIQUE, NOT NULL	Engine number
Registered_Date	DATE	NOT NULL	Date of vehicle registration

3.5 BIKE_COLOUR Table

Table 7:bike colour

Column	Data Type	Constraint	Description
Registration_No	VARCHAR(15)	PRIMARY KEY, FK → MOTORBIKE(Registration_No)	Links to motorbike

Colour	VARCHAR(30)	PRIMARY KEY	Colour applied to the bike
--------	-------------	-------------	----------------------------

This table resolves the multi-valued attribute 'colours' of MOTORBIKE. A motorbike with three colours would have three rows in this table.

3.6 ASSIGNMENT Table

Table 8:Assignment table

Column	Data Type	Constraint	Description
Assignment_ID	INT	PRIMARY KEY, AUTO_INCREMENT	Unique assignment identifier
Employee_No	VARCHAR(10)	NOT NULL, FK → RIDER(Employee_No)	Rider assigned
Registration_No	VARCHAR(15)	NOT NULL, FK → MOTORBIKE(Registration_No)	Motorbike used
Assignment_Date	DATE	NOT NULL	Date of assignment (shift date)
Start_Meter_Reading	DECIMAL(10,2)	NOT NULL	Odometer reading at assignment
End_Meter_Reading	DECIMAL(10,2)	NULL	Odometer reading when bike returned

3.7 ORDER Table

Table 9:order table

Column	Data Type	Constraint	Description
Order_No	VARCHAR(10)	PRIMARY KEY	Unique order number
Order_Date	DATE	NOT NULL	Date order was placed
Order_Time	TIME	NOT NULL	Time order was placed
Order_Status	VARCHAR(20)	NOT NULL	Status: Pending/Dispatched/Delivered

Payment_Method	VARCHAR(20)	NOT NULL	Cash, Card, Online, etc.
Order_Amount	DECIMAL(10,2)	NOT NULL	Total order value
Dispatch_Time	DATETIME	NULL	Timestamp when order dispatched
Customer_ID	VARCHAR(10)	NOT NULL, FK → CUSTOMER(Customer_ID)	Customer who placed the order
Employee_No	VARCHAR(10)	NULL, FK → RIDER(Employee_No)	Rider assigned to deliver

3.8 FOOD_ITEM Table

Table 10: food item table

Column	Data Type	Constraint	Description
Item_Number	VARCHAR(10)	PRIMARY KEY	Unique food item identifier
Item_Name	VARCHAR(100)	NOT NULL	Name of the food item
Item_Category	VARCHAR(50)	NOT NULL	Category (e.g. Burger, Pizza)
Price	DECIMAL(8,2)	NOT NULL	Unit price of the item

3.9 ORDER_ITEM Table

Table 11: order item table

Column	Data Type	Constraint	Description
Order_No	VARCHAR(10)	PRIMARY KEY, FK → ORDER(Order_No)	Links to order
Item_Number	VARCHAR(10)	PRIMARY KEY, FK → FOOD_ITEM(Item_Number)	Links to food item
Quantity	INT	NOT NULL, CHECK(Quantity > 0)	Number of this item ordered

3.10 INGREDIENT Table

Table 12: Ingredient table

Column	Data Type	Constraint	Description
Ingredient_ID	VARCHAR(10)	PRIMARY KEY	Unique ingredient identifier
Ingredient_Name	VARCHAR(100)	NOT NULL, UNIQUE	Name of the ingredient

3.11 ITEM_INGREDIENT Table

Column	Data Type	Constraint	Description
Item_Number	VARCHAR(10)	PRIMARY KEY, FK → FOOD_ITEM(Item_Number)	Links to food item
Ingredient_ID	VARCHAR(10)	PRIMARY KEY, FK → INGREDIENT(Ingredient_ID)	Links to ingredient

4. Normalisation

Normalisation is the process of organising a relational database to reduce data redundancy and improve data integrity. The process follows the normal forms: 1NF, 2NF, and 3NF.

4.1 First Normal Form (1NF)

1NF requires that: (1) each column contains atomic (indivisible) values, (2) each column contains values of a single type, (3) each column has a unique name, and (4) the order of rows does not matter.

Issues identified before 1NF:

- CUSTOMER address was a composite attribute (Location_No, Lane, Street, City) - resolved by storing each as a separate column.
- RIDER name was composite (First_Name, Middle_Name, Last_Name) - resolved by storing each part separately.
- MOTORBIKE had a multi-valued attribute (Colours) - resolved by creating the BIKE_COLOUR table.
- ORDER contained a repeating group of food items - resolved by creating the ORDER_ITEM junction table.
- FOOD_ITEM had a repeating group of ingredients - resolved by creating the ITEM_INGREDIENT junction table.
-

After 1NF: All tables contain only atomic values and no repeating groups.

4.2 Second Normal Form (2NF)

2NF requires that the table is in 1NF AND every non-key attribute is fully functionally dependent on the entire primary key (no partial dependencies). This only applies to tables with composite primary keys.

Issues identified:

- In ORDER_ITEM, if Item_Name or Price were stored here, they would be partially dependent on Item_Number alone (not the full composite PK). Resolution: Item_Name and Price remain only in FOOD_ITEM.
- In ITEM_INGREDIENT, no non-key attributes exist beyond the composite PK itself, so 2NF is satisfied by default.
- In DEPENDENT, Relationship is fully dependent on the composite PK (Employee_No + Dependent_Name + Date_of_Birth), so 2NF is satisfied.

After 2NF: No partial dependencies exist in any table.

4.3 Third Normal Form (3NF)

3NF requires that the table is in 2NF AND there are no transitive dependencies (non-key attributes must not depend on other non-key attributes).

Issues identified:

- In RIDER, 'Age' is transitively dependent on 'Date_of_Birth' (Age can be derived from DOB). Ideally Age should be computed dynamically; however, since the scenario explicitly states Age as an attribute, it is retained as a stored value with a note that it requires periodic update or trigger-based update.
- In ORDER, Order_Amount could be derived from the sum of (Quantity × Price) in ORDER_ITEM. It is retained as a stored value for performance reasons and to capture the historical price at time of order.
- No other transitive dependencies were found across the schema.

After 3NF: The database design is fully normalised to 3NF. The final schema contains 11 interrelated tables, satisfying the minimum requirement of 6 tables.

5. Database Schema

5.1 Textual Format (SQL DDL Statements)

```
CREATE TABLE CUSTOMER (
  Customer_ID   VARCHAR(10) PRIMARY KEY,
  Customer_Name VARCHAR(100) NOT NULL,
  NIC           VARCHAR(12) UNIQUE NOT NULL,
  Date_of_Birth DATE      NOT NULL,
  Contact_Number VARCHAR(15) NOT NULL,
  Location_No   VARCHAR(10) NOT NULL,
  Lane          VARCHAR(50) NULL,
  Street        VARCHAR(100) NOT NULL,
  City          VARCHAR (50) NOT NULL );
```

```
CREATE TABLE RIDER (
  Employee_No   VARCHAR(10) PRIMARY KEY,
  First_Name    VARCHAR(50) NOT NULL,
  Middle_Name   VARCHAR(50) NULL,
  Last_Name     VARCHAR(50) NOT NULL,
  NIC           VARCHAR(12) UNIQUE NOT NULL,
  Date_of_Birth DATE      NOT NULL,
```

```

Contact_Number VARCHAR(15) NOT NULL,
License_Number VARCHAR(20) UNIQUE NOT NULL,
Address        VARCHAR(200) NOT NULL,
Age            INT          NOT NULL );
CREATE TABLE DEPENDENT (
Employee_No    VARCHAR(10) NOT NULL,
Dependent_Name VARCHAR(100) NOT NULL,
Date_of_Birth  DATE        NOT NULL,
Relationship    VARCHAR(50) NOT NULL,
PRIMARY KEY (Employee_No, Dependent_Name, Date_of_Birth),
FOREIGN KEY (Employee_No) REFERENCES RIDER(Employee_No) );
CREATE TABLE MOTORBIKE (
Registration_No VARCHAR(15) PRIMARY KEY,
Brand           VARCHAR(50) NOT NULL,
Model           VARCHAR(50) NOT NULL,
Engine_Number   VARCHAR(20) UNIQUE NOT NULL,
Registered_Date DATE        NOT NULL );
CREATE TABLE BIKE_COLOUR (
Registration_No VARCHAR(15) NOT NULL,
Colour          VARCHAR(30) NOT NULL,
PRIMARY KEY (Registration_No, Colour),
FOREIGN KEY (Registration_No) REFERENCES MOTORBIKE(Registration_No) );
CREATE TABLE FOOD_ITEM (
Item_Number     VARCHAR(10) PRIMARY KEY,
Item_Name       VARCHAR(100) NOT NULL,
Item_Category   VARCHAR(50) NOT NULL,
Price           DECIMAL(8,2) NOT NULL );
CREATE TABLE INGREDIENT (
Ingredient_ID    VARCHAR(10) PRIMARY KEY,
Ingredient_Name  VARCHAR(100) UNIQUE NOT NULL );
CREATE TABLE ITEM_INGREDIENT (
Item_Number     VARCHAR(10) NOT NULL,
Ingredient_ID    VARCHAR(10) NOT NULL,
PRIMARY KEY (Item_Number, Ingredient_ID),
FOREIGN KEY (Item_Number) REFERENCES FOOD_ITEM(Item_Number),
FOREIGN KEY (Ingredient_ID) REFERENCES INGREDIENT(Ingredient_ID) );
CREATE TABLE `ORDER` (
Order_No        VARCHAR(10) PRIMARY KEY,
Order_Date      DATE        NOT NULL,
Order_Time      TIME        NOT NULL,
Order_Status    VARCHAR(20) NOT NULL,
Payment_Method  VARCHAR(20) NOT NULL,

```

```

Order_Amount    DECIMAL(10,2) NOT NULL,
Dispatch_Time   DATETIME    NULL,
Customer_ID     VARCHAR(10)  NOT NULL,
Employee_No     VARCHAR(10)  NULL,
FOREIGN KEY (Customer_ID) REFERENCES CUSTOMER(Customer_ID),
FOREIGN KEY (Employee_No) REFERENCES RIDER(Employee_No) );

CREATE TABLE ORDER_ITEM (
    Order_No      VARCHAR(10) NOT NULL,
    Item_Number   VARCHAR(10) NOT NULL,
    Quantity      INT        NOT NULL CHECK(Quantity > 0),
    PRIMARY KEY (Order_No, Item_Number),
    FOREIGN KEY (Order_No) REFERENCES `ORDER`(Order_No),
    FOREIGN KEY (Item_Number) REFERENCES FOOD_ITEM(Item_Number) );

CREATE TABLE ASSIGNMENT (
    Assignment_ID INT        PRIMARY KEY AUTO_INCREMENT,
    Employee_No   VARCHAR(10) NOT NULL,
    Registration_No VARCHAR(15) NOT NULL,
    Assignment_Date DATE      NOT NULL,
    Start_Meter_Reading DECIMAL(10,2) NOT NULL,
    End_Meter_Reading  DECIMAL(10,2) NULL,
    FOREIGN KEY (Employee_No) REFERENCES RIDER(Employee_No),
    FOREIGN KEY (Registration_No) REFERENCES MOTORBIKE(Registration_No) );

```

5.2 Graphical Format – Schema Diagram Description

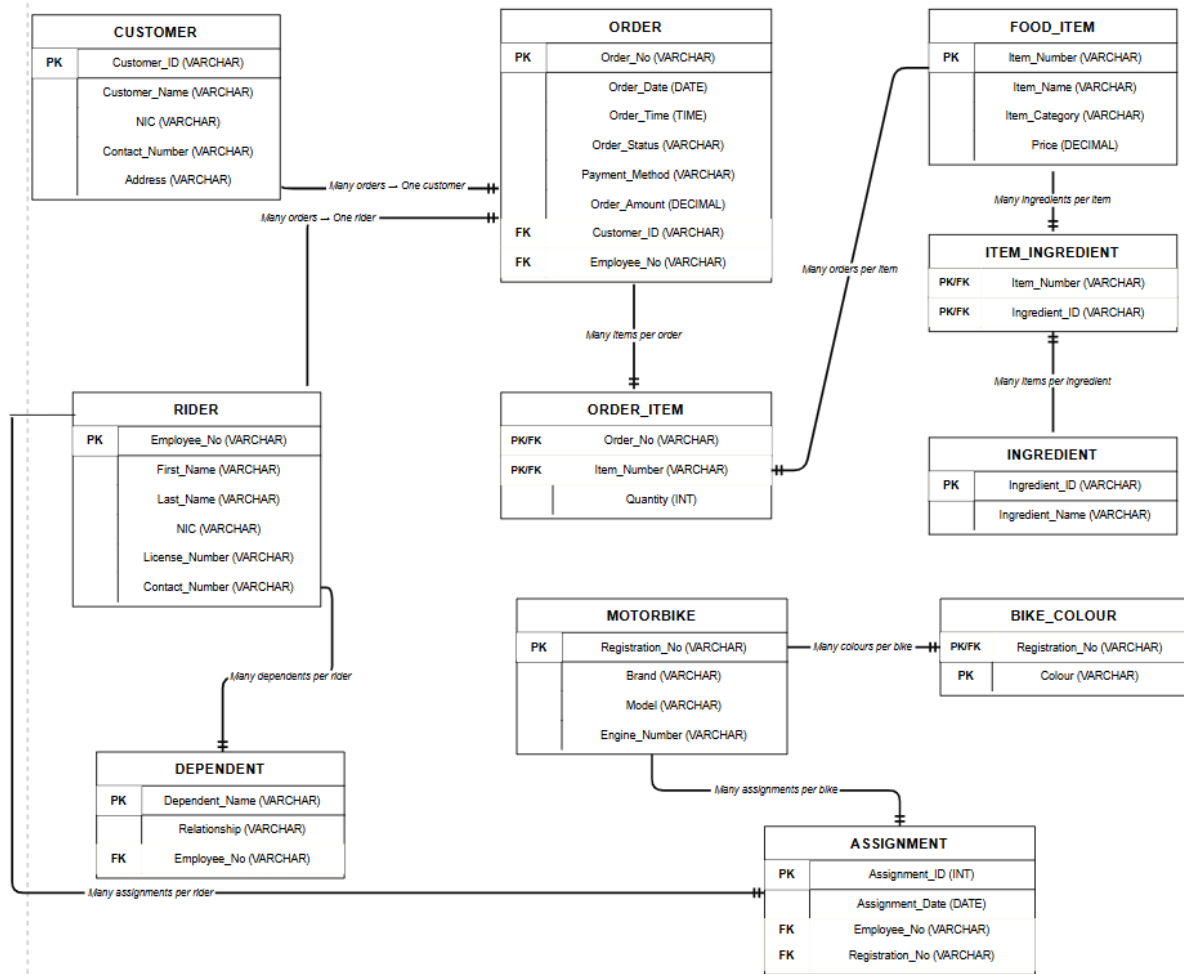


Figure 2:schema diagram

Table	Connected To	FK Column	Relationship
Table 13:schema table ORDER	CUSTOMER	Customer_ID	Many orders → One customer
ORDER	RIDER	Employee_No	Many orders → One rider
ORDER_ITEM	ORDER	Order_No	Many items per order
ORDER_ITEM	FOOD_ITEM	Item_Number	Many orders per item
ITEM_INGREDIENT	FOOD_ITEM	Item_Number	Many ingredients per item
ITEM_INGREDIENT	INGREDIENT	Ingredient_ID	Many items per ingredient
DEPENDENT	RIDER	Employee_No	Many dependents per rider

BIKE_COLOUR	MOTORBIKE	Registration_No	Many colours per bike
ASSIGNMENT	RIDER	Employee_No	Many assignments per rider
ASSIGNMENT	MOTORBIKE	Registration_No	Many assignments per bike

6. Interface Wireframe Designs

The following wireframe descriptions represent simple interface designs for the FoodHub system. Each wireframe represents either an input form or an output screen.

6.1 Wireframe 1 – Customer Registration Form (Input)

Customer Registration Form

✓ Record saved successfully | Customer ID: C-0041

CUSTOMER DETAILS

Customer ID: Auto-generated

Customer Name *: Enter full name

NIC *: National ID number

Date of Birth *: DD / MM / YYYY

Contact Number *: +94 ____

ADDRESS (DELIVERY ADDRESS)

Location No. *: No.

Lane: Lane name

Street *: Street name

City *: Select city ▾

Save Customer Clear Form Cancel

Figure 3: Customer Registration Form

Purpose: Allows staff to register a new customer into the system.

Components:

- Form title: 'New Customer Registration'
- Text fields: Customer ID (auto-generated), Customer Name, NIC, Date of Birth (date picker), Contact Number
- Address section: Location No, Lane, Street, City (dropdown of major cities)
- Buttons: [Save Customer] [Clear Form] [Cancel]
- Output: Success/Error message label at top of form

6.2 Wireframe 2 – Place New Order Form (Input)

Place New Order Form

ORDER HEADER

Order No. Order Date Order Time

Customer ID *

ORDER ITEMS

Item No.	Item Name	Category	Price	Qty	Subtotal
ITM-001	Chicken Burger	Burgers	\$5.50	2	\$11.00
ITM-004	French Fries	Sides	\$2.00	1	\$2.00

ORDER SUMMARY

Payment Method Order Status

Subtotal: \$13.00
Total: \$13.00

Figure 4: place new order

Purpose: Allows staff to record a new food order placed by a customer over the phone.

Components:

- Order header: Order No (auto-generated), Order Date (auto), Order Time (auto), Customer ID (lookup field with search button)
- Customer details display (read-only): Name, Contact, Address (populated after customer lookup)
- Order items section: Table with columns [Item No | Item Name | Category | Price | Qty | Subtotal]
- Add Item button: opens item search popup

- Order summary: Payment Method (dropdown), Order Amount (calculated), Order Status (default: Pending)
- Buttons: [Place Order] [Clear] [View Ingredients] [Cancel]

6.3 Wireframe 3 – Order Dispatch Screen (Input)

Order Dispatch Screen

ORDER LOOKUP

ORD-2025-0318 · Nimal Perera · 12 Galle Rd, Colombo 3
Items: Chicken Burger x2, French Fries x1 · Total: \$13.00 · Status: Pending

RIDER & BIKE ASSIGNMENT

Rider (Employee No.) *

Motorbike Registration *

DISPATCH DETAILS

Dispatch Time

Start Meter Reading (km) *

Figure 5: Order Dispatch Screen

Purpose: Records the dispatch time and assigns a rider when an order is dispatched.

Components:

- Search: Order No text field + [Search] button
- Order details display (read-only): Customer, Address, Items, Amount
- Rider assignment: Employee No (dropdown of available riders), Motorbike Registration (dropdown)
- Dispatch Time: auto-populated with current datetime (editable)
- Meter readings: Start Meter Reading text field
- Buttons: [Dispatch Order] [Cancel]

6.4 Wireframe 4 – Order Inquiry Screen (Output)

Order Inquiry Screen (Output)

Search by Customer ID or Order No. Search Clear

Order No.	Date	Time	Status	Amount	Rider
ORD-2025-0318	16/03/2025	14:22	Dispatched	\$13.00	EMP-012
ORD-2025-0305	05/03/2025	11:10	Delivered	\$8.50	EMP-007

ORDER DETAIL — ORD-2025-0318

Item Name	Qty	Subtotal	Ingredients
Chicken Burger	2	\$11.00	Bun, Chicken Patty, Lettuce, Tomato, Sauce, Cheese
French Fries	1	\$2.00	Potato, Oil, Salt, Seasoning

Print Back

Figure 6:Order Inquiry Screen (Output)

Purpose: Staff can view all orders for a customer or view order details including ingredients.

Components:

- Search bar: Search by Customer ID or Order No + [Search] button
- Orders table (output): columns [Order No | Date | Time | Status | Amount | Rider]
- Select order → View detail panel: Items ordered, Ingredients per item (expandable)
- Buttons: [Print] [Back]

6.5 Wireframe 5 – Rider Registration Form (Input)

Rider Registration Form

RIDER DETAILS

Employee No.
Auto-generated

First Name *
First name

Middle Name
Optional

Last Name *
Last name

NIC *
National ID

Date of Birth *
DD / MM / YYYY

Age
Auto

Contact No. *
+94 _____

License Number *
License no.

Address *
Full address

DEPENDENTS

Dependent Name	Relationship	Date of Birth	Remove
Kamala Silva	Spouse	12/04/1990	[Remove]
Amal Silva	Child	03/07/2015	[Remove]
+ Add Dependent			

Save Rider

Clear

Cancel

Figure 7:Rider Registration Form (Input)

Purpose: Record details of a new rider/employee.

Components:

- Text fields: Employee No (auto), First Name, Middle Name, Last Name, NIC, Date of Birth, Contact Number, License Number, Address, Age
- Dependents section: sub-table [Dependent Name | Relationship | Date of Birth] with [Add Row] [Remove Row] buttons
- Buttons: [Save Rider] [Clear] [Cancel]

7. Flowcharts for Major Processes

Flowcharts use the following standard symbols: Oval (Start/End), Rectangle (Process), Diamond (Decision), Parallelogram (Input/Output), Arrow (Flow direction). Draw these using Lucidchart, draw.io, or Microsoft Visio.

7.1 Flowchart 1 – Place Order Process

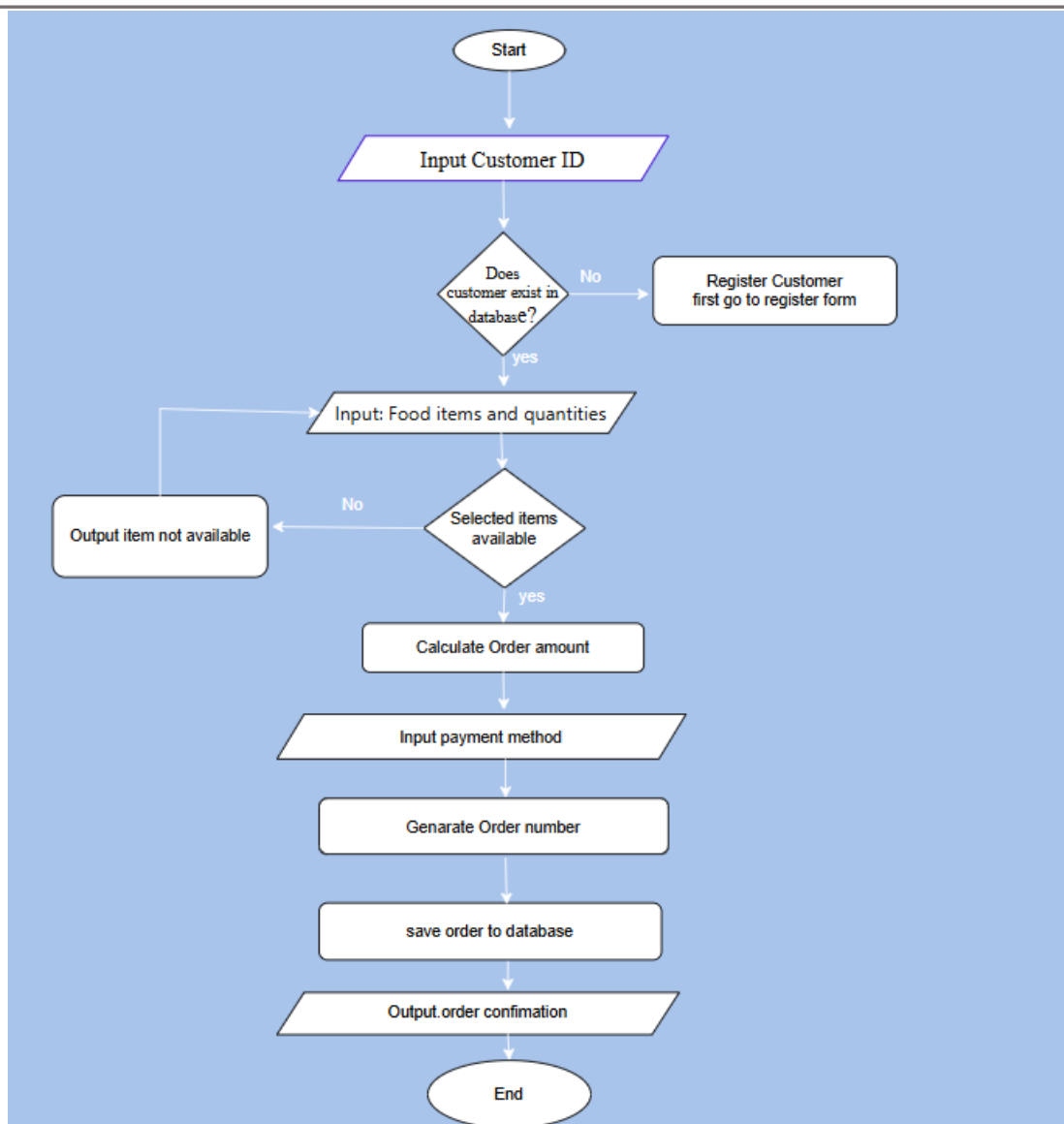


Figure 8:Place Order Process

The Place Order Process flowchart illustrates the sequential steps followed by a staff member when recording a new food order placed by a customer over the telephone. The process begins when a customer calls to place an order. The staff member first verifies whether the customer exists in the system; if not, a new customer record is created. The staff member then selects the required food items and quantities, and the system calculates the total order amount. Payment method is confirmed, the order status is set to Pending, and the order record is saved to the database. The process ends with an order confirmation being provided to the customer.

7.2 Flowchart 2 – Order Dispatch Process

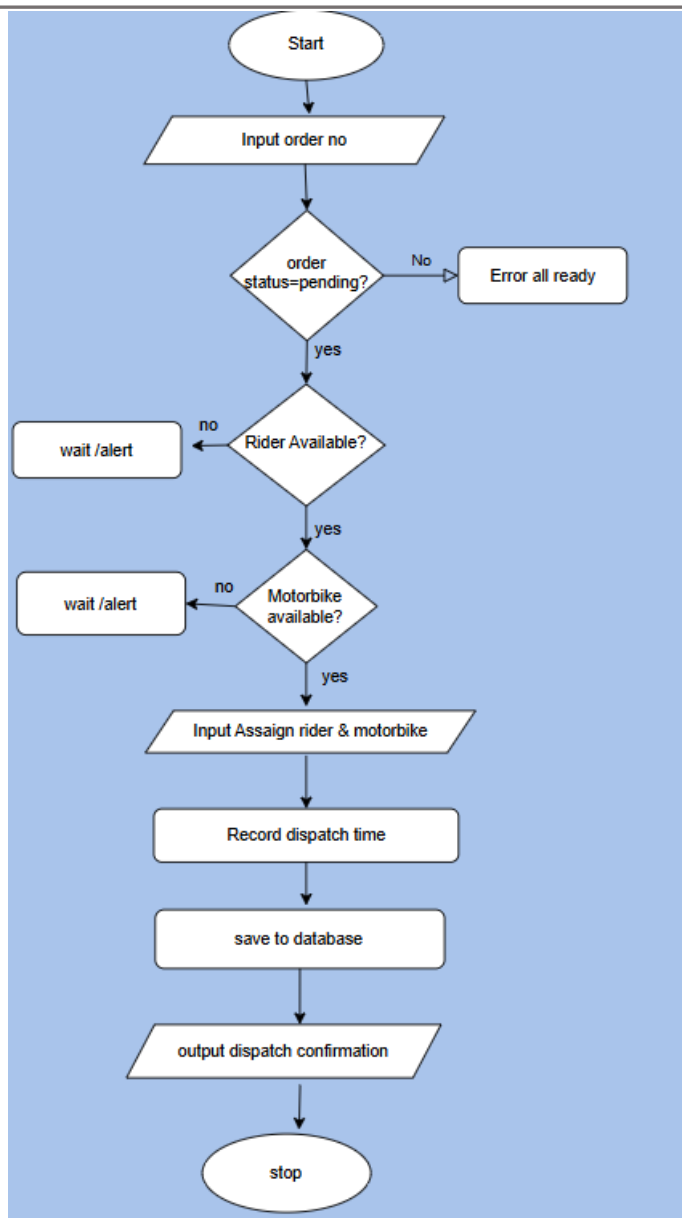


Figure 9:Order Dispatch Process

The Order Dispatch Process flowchart demonstrates the steps taken by dispatch staff when an order is ready to be delivered to a customer. The process begins by searching for a pending order using the Order Number. The dispatch staff then checks whether an available rider exists for the current shift. If a rider is available, an available motorbike is also identified and assigned to the rider. The start meter reading of the assigned motorbike is recorded in the system. The dispatch time is automatically captured and the order status is updated to Dispatched. The process concludes once the assignment record is saved and the rider departs with the order for delivery.

7.3 Flowchart 3 – Customer Registration Process

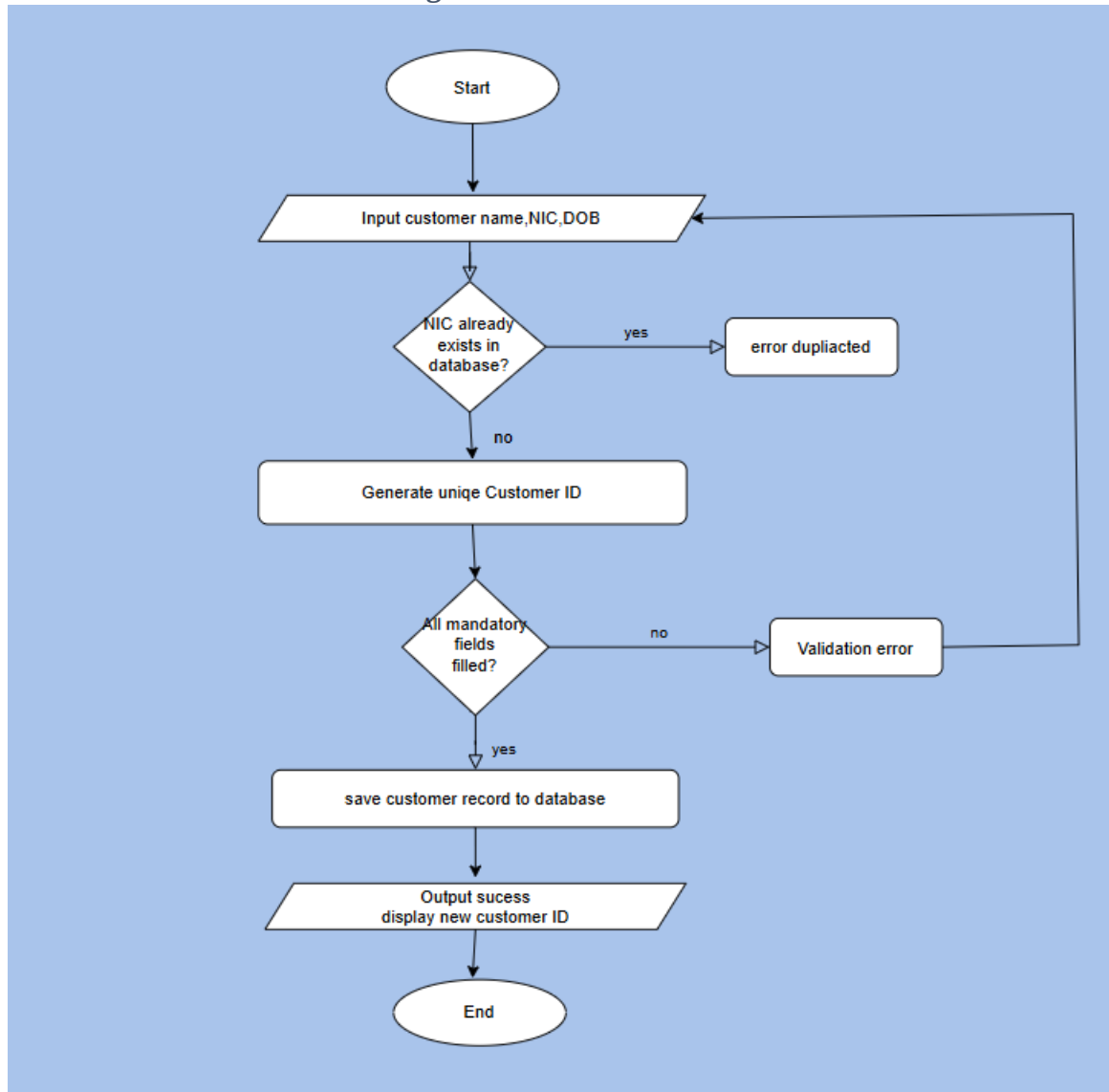


Figure 10:Customer Registration Process

The Customer Registration Process flowchart outlines the steps followed by a staff member when registering a new customer into the FoodHub system. The process is initiated when a new customer contacts the company to place their first order. The staff member collects the customer's personal details including their name, NIC, date of birth, contact number, and delivery address comprising Location Number, Lane, Street, and City. The system checks whether a customer with the same NIC already exists to prevent duplicate records. If the customer is new, a unique Customer ID is automatically generated and the record is saved to the CUSTOMER table. The process ends with the new customer record successfully stored in the database, ready for order placement.

7.4 Flowchart 4 – Bike Assignment to Rider (Shift Start)

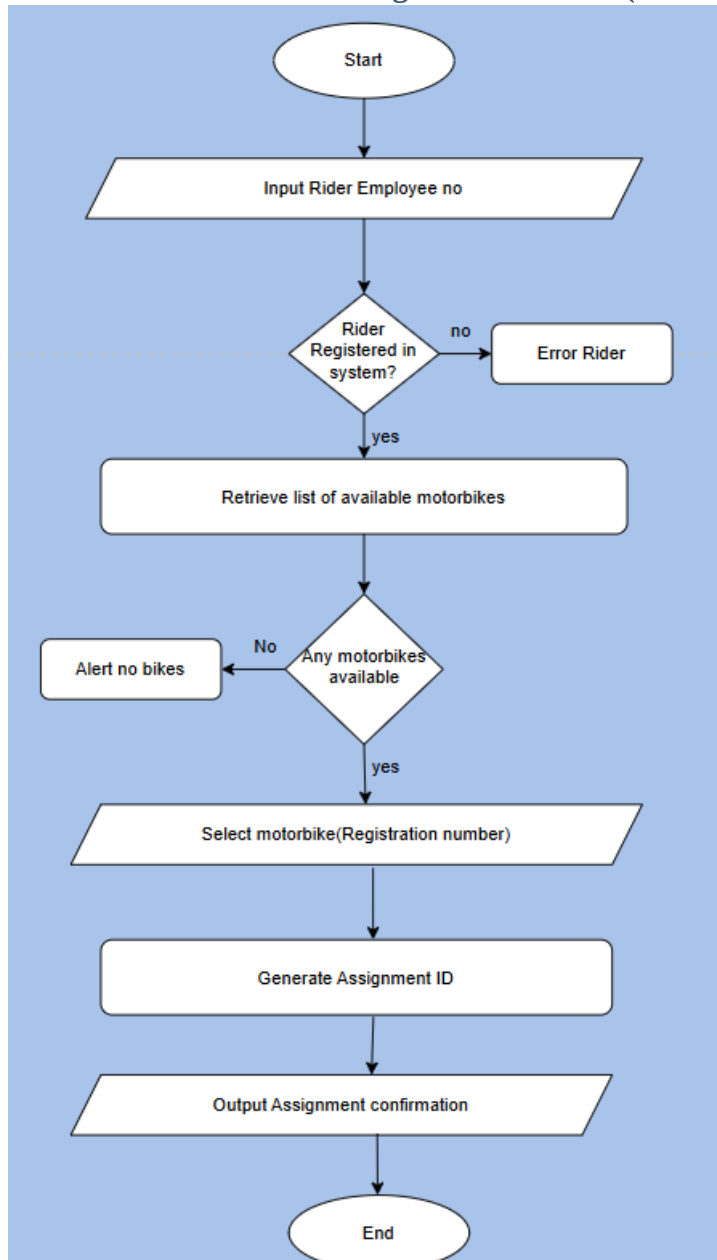


Figure 11: Bike Assignment to Rider (Shift Start)

The Bike Assignment to Rider flowchart describes the process of assigning an available company motorbike to a rider at the start of their shift. The process begins when a rider reports for duty. The system checks the rider's Employee Number to confirm their identity and availability. The dispatch staff then checks the list of available motorbikes that are not currently assigned to any other rider. An available motorbike is selected and its Registration Number is recorded. The start meter reading is captured to track the distance travelled during the shift. A new Assignment record is created in the ASSIGNMENT table linking the rider and the motorbike with the assignment date and start meter reading. The process concludes when the assignment is confirmed and the rider departs for deliveries.

8. Use Case Diagram

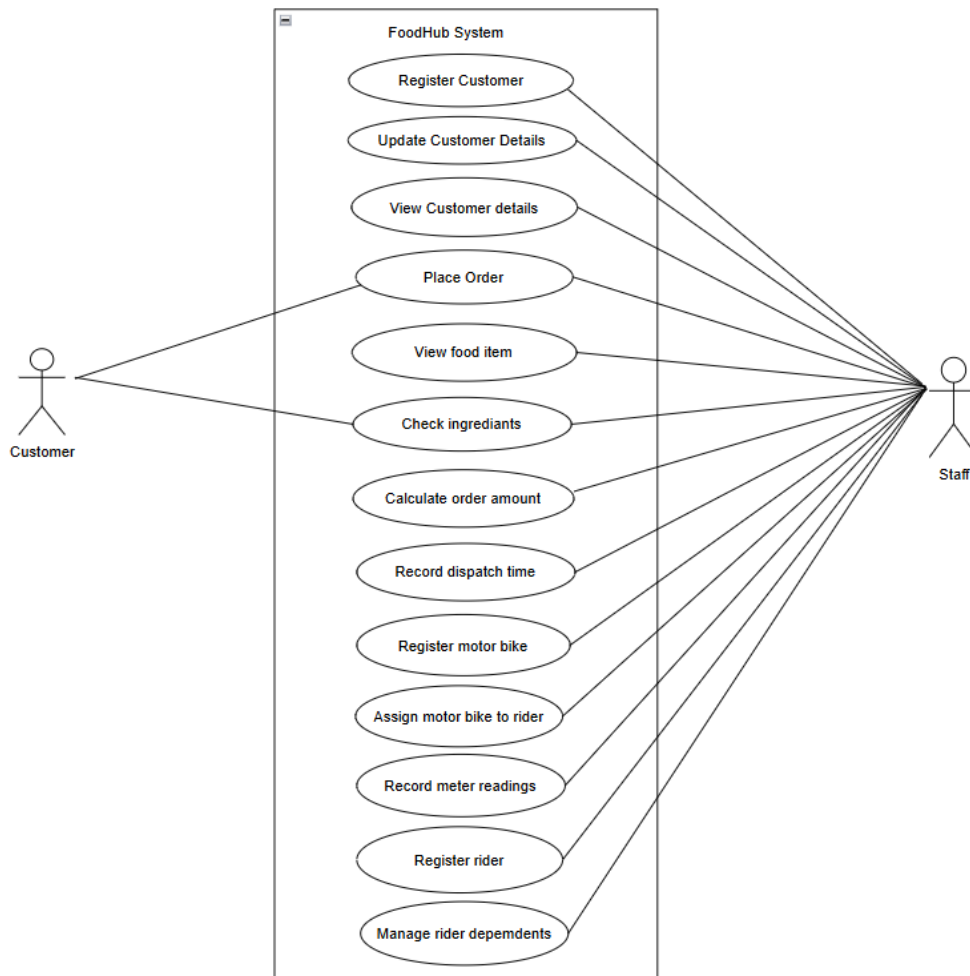


Figure 12:Bike Assignment to Rider (Shift Start)

8.1 Actors

Table 14:actor

Actor	Role Description
Staff Member	Records orders, registers customers, provides ingredient information to customers
Dispatch Staff	Records dispatch time, assigns riders and motorbikes to orders
Rider (Employee)	Delivers food orders to customers
System Administrator	Manages system configurations, motorbikes, food items, ingredients
Customer (External)	Places orders by phone; receives delivery

8.2 Use Cases

Table 15: usecases

Use Case	Actor(s)	Description
UC01 – Register Customer	Staff Member	Create new customer record in the system
UC02 – Update Customer	Staff Member	Modify existing customer details
UC03 – Place Order	Staff Member	Record a new food order for a customer
UC04 – Add Order Items	Staff Member	Add food items and quantities to an order (included in UC03)
UC05 – View Ingredients	Staff Member	Retrieve ingredient list for a food item during ordering
UC06 – Dispatch Order	Dispatch Staff	Record dispatch time and assign rider + bike to order
UC07 – Assign Motorbike	Dispatch Staff	Assign an available motorbike to a rider for a shift
UC08 – Record Meter Reading	Dispatch Staff	Record start and end meter readings for bike assignment
UC09 – Return Motorbike	Dispatch Staff / Rider	Record end meter reading when bike is returned
UC10 – Register Rider	System Administrator	Add a new rider/employee to the system
UC11 – Add Dependents	System Administrator	Record dependent information for a rider
UC12 – Add Food Item	System Administrator	Add new food items to the menu
UC13 – Manage Ingredients	System Administrator	Add or update ingredient records
UC14 – Add Motorbike	System Administrator	Register a new motorbike in the system
UC15 – View Order History	Staff Member	View orders placed by a specific customer
UC16 – Update Order Status	Dispatch Staff	Update status to Delivered when delivery confirmed

Relationships in the Use Case Diagram:

- UC04 (Add Order Items) has an <<include>> relationship with UC03 (Place Order) — adding items is mandatory when placing an order.
- UC05 (View Ingredients) has an <<extend>> relationship with UC03 (Place Order) — viewing ingredients is optional during ordering.
- UC07 (Assign Motorbike) has an <<include>> relationship with UC06 (Dispatch Order).
- UC08 (Record Meter Reading) has an <<include>> relationship with UC07 (Assign Motorbike).
- UC11 (Add Dependents) has an <<extend>> relationship with UC10 (Register Rider).

9. Class Diagram

The class diagram models the system's object-oriented structure corresponding to the database entities. No access level modifiers are shown as per the assignment requirement.

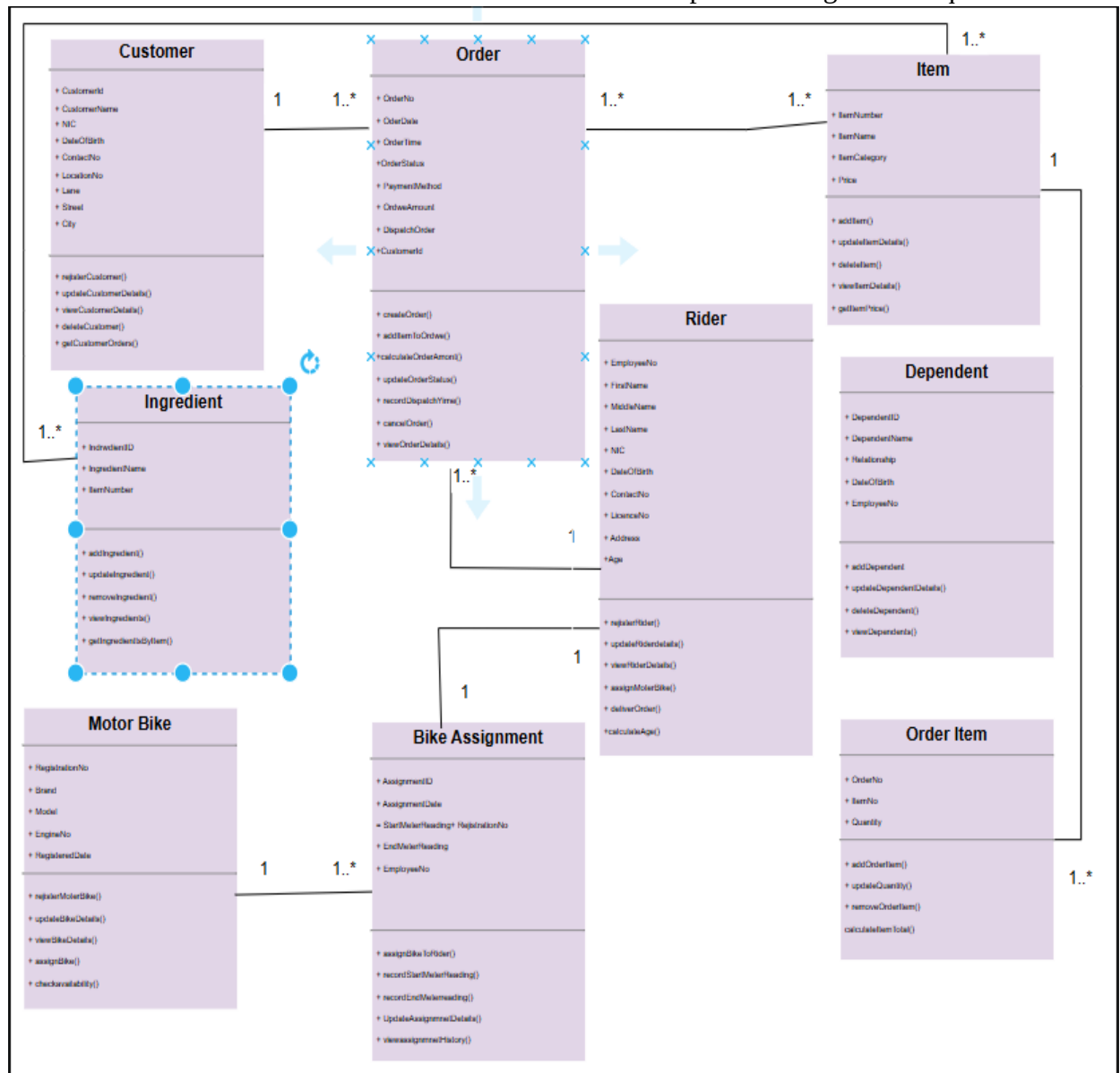


Figure 13: class diagram

Table 16: class diagram

Class	Attributes	Methods
Customer	customerId, customerName, nic, dateOfBirth, contactNumber, locationNo, lane, street, city	registerCustomer(), updateCustomer(), getOrderHistory()
Order	orderNo, orderDate, orderTime, orderStatus, paymentMethod, orderAmount, dispatchTime, customerId, employeeNo	placeOrder(), dispatchOrder(), updateStatus(), calculateAmount()
FoodItem	itemNumber, itemName, itemCategory, price	addItem(), updateItem(), getIngredients()
Ingredient	ingredientId, ingredientName	addIngredient(), updateIngredient()
Rider	employeeNo, firstName, middleName, lastName, nic, dateOfBirth, contactNumber, licenseNumber, address, age	registerRider(), updateRider(), addDependent(), getAssignmentHistory()
Dependent	employeeNo, dependentName, dateOfBirth, relationship	addDependent(), updateDependent()
Motorbike	registrationNo, brand, model, engineNumber, registeredDate, colours	addBike(), updateBike(), getColours()
Assignment	assignmentId, employeeNo, registrationNo, assignmentDate, startMeterReading, endMeterReading	createAssignment(), recordReturn(), getAssignmentsByDate()
OrderItem	orderNo, itemNumber, quantity	addOrderItem(), removeOrderItem()

Class Relationships (to depict in diagram):

- Customer 1 — * Order (Association: One customer places many orders)

- Order * — * FoodItem (Association via OrderItem: Many-to-many)
- FoodItem * — * Ingredient (Association via ItemIngredient: Many-to-many)
- Rider 1 — * Dependent (Composition: Dependent cannot exist without Rider)
- Rider * — * Motorbike (Association via Assignment: Many-to-many)
- Order 1 — * OrderItem (Composition)
- Motorbike 1 — * BikeColour (Composition)

10. Data Flow Diagrams (Gane & Sarson Notation)

The DFDs are drawn using Gane & Sarson notation: Rectangle with rounded corners = External Entity, Rounded rectangle = Process, Open-ended rectangle = Data Store, Arrow = Data Flow. Draw these using Lucidchart, draw.io, or Visio.

10.1 Context Level DFD (Level 0)

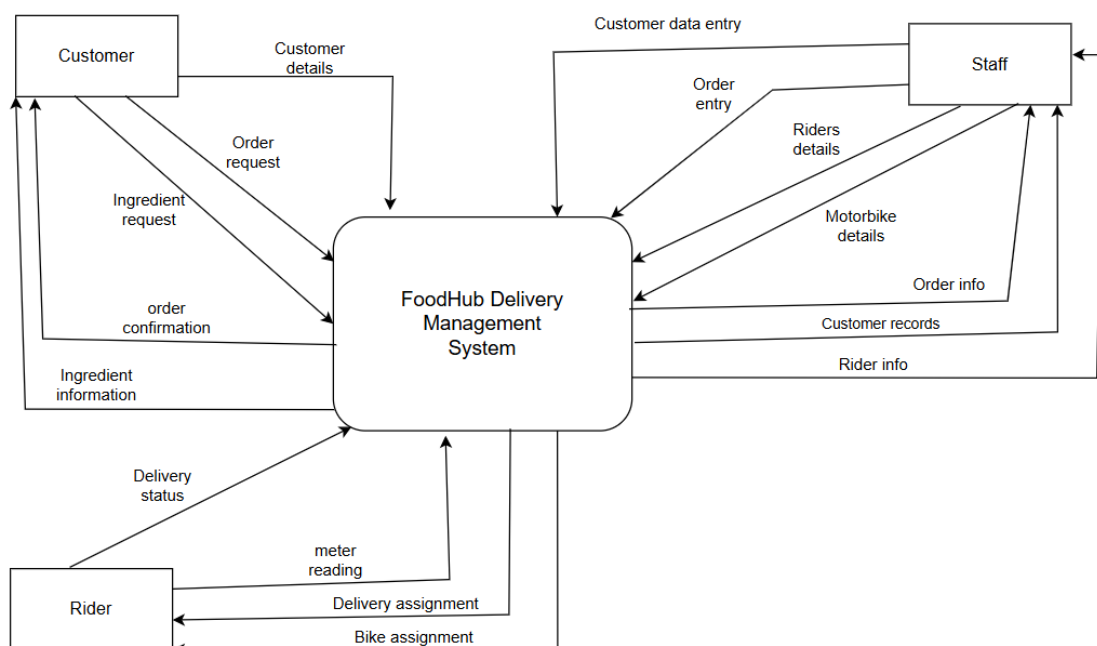


Figure 14: context diagram

The context level shows the system as a single process (Process 0) with all external entities and their data flows.

External Entities:

- Customer – provides order requests, receives order confirmation and food delivery
- Rider – receives delivery assignment, provides meter readings
- System Administrator – provides system configuration data (menu items, bikes, ingredients)

Data Flows into System:

- Customer → System: Customer details, Order request, Item selections, Payment method
- Rider → System: Meter reading (return), Delivery confirmation
- Admin → System: Food item data, Ingredient data, Motorbike data, Rider data

Data Flows out of System:

- System → Customer: Order confirmation, Ingredient information
- System → Rider: Delivery assignment details
- System → Admin: Reports (future module), System confirmations

10.2 Level 1 DFD – Decomposed Processes

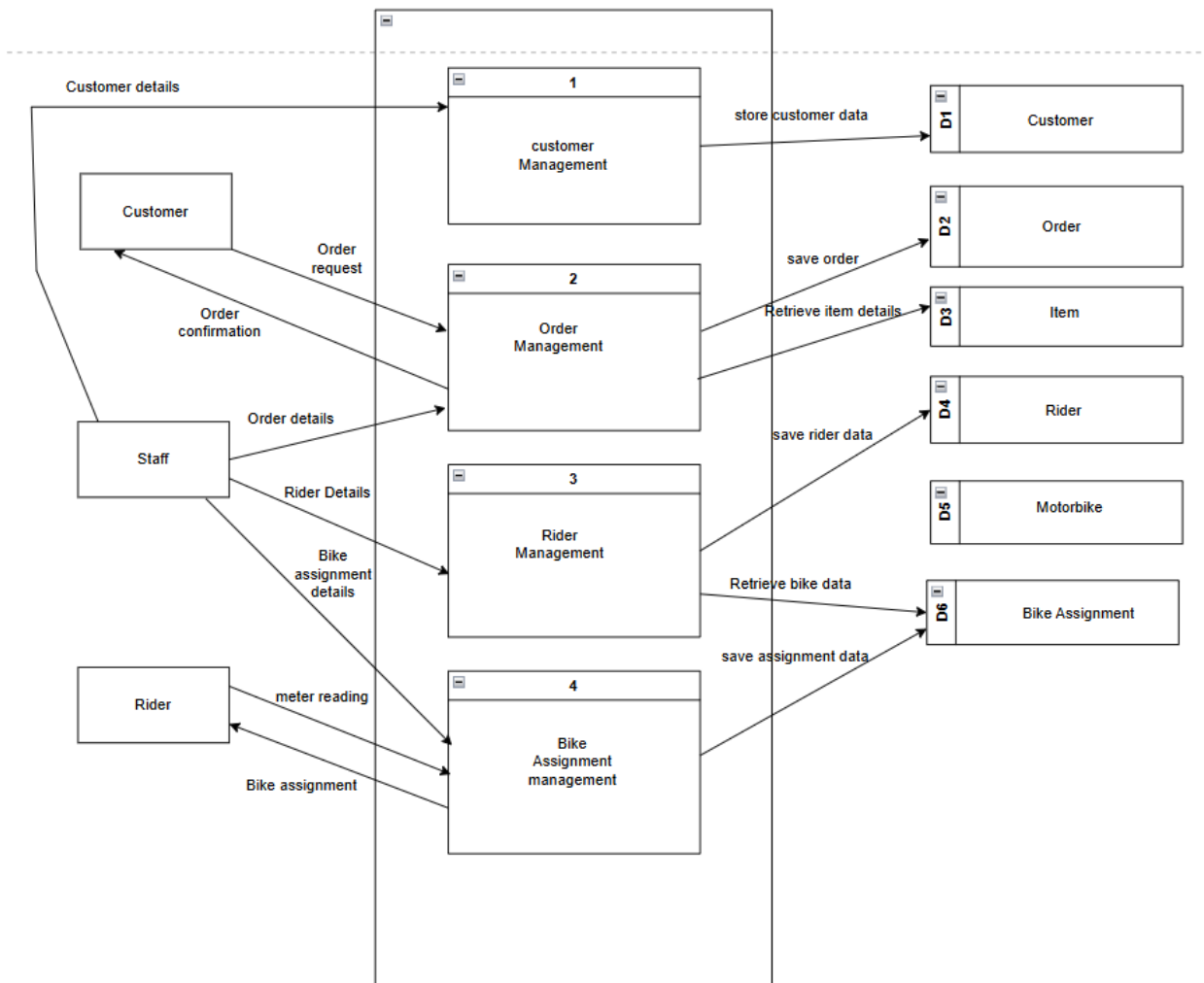


Figure 15:dfd level 1

Level 1 decomposes the single context process into the main sub-processes with their data stores.

Table 17:lever 1 dfd

Process No	Process Name	Inputs	Outputs	Data Stores Used
2.0	Process Order	Order details, Customer	Order No, Confirmation, Ingredient info	D1: CUSTOMER D2: ORDER D3:FOOD_ITEM D4: ORDER_ITEM

		ID, Item selections		
3.0	Dispatch Order	Order No, Rider, Bike, Dispatch time, Meter reading	Dispatch confirmation, Assignment record	D2: ORDER D5: ASSIGNMENT D6: MOTORBIKE
4.0	Manage Riders	Rider details, Dependent details	Employee No, Confirmation	D7:RIDER D8: DEPENDENT
5.0	Manage Motorbikes	Bike details, Colour data	Registration No, Confirmation	D6:MOTORBIKE D9: BIKE_COLOUR
6.0	Manage Menu	Food item data, Ingredient data	Item No, Ingredient ID, Confirmation	D3:FOOD_ITEM D10:INGREDIENT D11: ITEM_INGREDIENT
7.0	Generate Reports (Future Module)	Query parameters	Reports (delivery history, rider performance)	All Data Stores

Data Stores Summary:

- D1: CUSTOMER – stores all customer records
- D2: ORDER – stores all order records
- D3: FOOD_ITEM – stores all menu item records
- D4: ORDER_ITEM – junction store for order items
- D5: ASSIGNMENT – stores bike assignment history
- D6: MOTORBIKE – stores motorbike records
- D7: RIDER – stores rider/employee records
- D8: DEPENDENT – stores rider dependent records
- D9: BIKE_COLOUR – stores bike colour data
- D10: INGREDIENT – stores ingredient records
- D11: ITEM_INGREDIENT – junction store for item ingredients

11. Evaluation of the Effectiveness of Designs

Each design is evaluated against the identified user and system requirements to assess its effectiveness.

11.1 ER Model Evaluation

The ER model effectively represents all entities and their relationships as described in the FoodHub scenario. All user requirements were mapped to entities: customers, orders, food items, ingredients, riders, dependents, and motorbikes are all captured. The many-to-many relationships between Orders and Items, and between Food Items and Ingredients, were correctly identified and modelled with junction entities. The weak entity pattern for

Dependents is the most appropriate solution for the missing unique identifier problem, directly addressing the requirement. The multi-valued attribute for motorbike colours was correctly handled by externalising it to BIKE_COLOUR. One limitation is that the ER diagram does not inherently capture the business rule that every food item must have at least three ingredients; this constraint must be enforced at the application level.

11.2 Logical Database Design Evaluation

The logical schema with 11 interrelated tables (exceeding the minimum requirement of 6) fully implements the ER model. All primary and foreign keys are explicitly defined, and referential integrity is maintained across all relationships. The schema correctly supports all user requirements: tracking which rider delivered which order, which bike was used on which date, the ingredients of each food item, and the dependents of each rider. The use of a composite primary key for the ASSIGNMENT identification and for DEPENDENT resolution demonstrates sound relational theory application. Storing Order_Amount as a persistent value (rather than computing it each time) is a justified denormalisation for performance and historical accuracy.

11.3 Normalisation Evaluation

The database has been normalised to Third Normal Form (3NF), eliminating insertion, update, and deletion anomalies. The 1NF conversion removed all repeating groups and multi-valued attributes. The 2NF step ensured that junction tables do not contain attributes partially dependent on the composite key. The 3NF analysis identified the transitive dependency of Age on Date_of_Birth; retaining Age as a stored value is an accepted practical trade-off explicitly noted in the design. The normalised design ensures data consistency: changing a food item's price requires updating only the FOOD_ITEM table, and this change will automatically be reflected in all new orders using that item.

11.4 Interface Wireframe Evaluation

The five wireframe designs cover the core input and output interfaces identified in the system requirements: customer registration (input), order placement (input/output), order dispatch (input), order inquiry (output), and rider registration (input). Each wireframe was designed to be intuitive for non-technical staff operating the system. The order placement form integrates customer lookup, item selection, and ingredient viewing in a single interface, reducing the number of screens a staff member must navigate during a phone call. Limitation: The wireframes are described textually here; in the actual submission they must be visually drawn. The designs do not yet include all future HR and Finance module screens, as these are outside the current scope.

11.5 Flowchart Evaluation

The four flowcharts cover the four most critical processes: placing an order, dispatching an order, registering a customer, and assigning a motorbike. Each flowchart includes appropriate decision points (existence checks, availability checks, validation checks) that reflect real business rules and prevent data integrity violations. For example, the order

placement flowchart checks customer existence before proceeding, and the dispatch flowchart checks both rider and bike availability. The flowcharts provide clear procedural guidance for developers implementing the system logic and for staff training purposes.

11.6 Use Case Diagram Evaluation

The use case diagram captures all 16 identified use cases across four actors, providing a comprehensive view of the system's functional scope from the user's perspective. The use of <<include>> and <<extend>> relationships correctly models mandatory and optional sub-processes (e.g., viewing ingredients is optional during order placement, while adding items is mandatory). The diagram clearly delineates responsibilities between Staff, Dispatch Staff, Riders, and Administrators, which is valuable for access control design in the next development phase. A limitation is that the diagram does not model reporting use cases in detail, as these are deferred to the next development phase.

11.7 Class Diagram Evaluation

The class diagram successfully maps the database entities to object-oriented classes, providing a foundation for the system's software architecture. All major classes include relevant attributes and meaningful methods reflecting the system's operations. The composition relationships (Rider-Dependent, Order-OrderItem) and association relationships are correctly modelled, giving developers a clear blueprint for implementing the business logic layer. The class diagram supports the future integration of HR and Finance modules by providing clear class boundaries and a modular structure.

11.8 Data Flow Diagram Evaluation

The context-level DFD correctly identifies all three external entities (Customer, Rider, Administrator) and summarises all data flows into and out of the system at a high level. The Level 1 DFD decomposes the system into seven meaningful processes, each with clearly defined inputs, outputs, and data stores. This two-level approach using Gane & Sarson notation provides both a strategic overview for stakeholders and a detailed process view for developers. The future Reports process (7.0) is included in the DFD design even though it is not implemented in the first development phase, satisfying the requirement that report generation processes be included in the design even before implementation. The 11 data stores in the Level 1 DFD correspond directly to the 11 database tables, demonstrating consistency across all design artefacts.

12. References

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